PRECIFARM
Modular Energy Platform

PF-MODENERGY-DESIGN-001
v1.5
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INDUSTRIAL DESIGN OVERVIEW · 5 PAGES

One module.
Three scales.

Conceptual industrial design for P1 Go, Energy Module, P2 Home and Pod — unified aluminum-and-glass language for Kenya homes and SMEs.

P1 GO | P2 HOME | POD | ENERGY OS | EV CHARGING

Render 1 — Product family. Seamless aluminum, glass UI strips, one repeated Energy Module. CONCEPTUAL.

DOCUMENT	STATUS	COMPANION
PF-MODENERGY-DESIGN-001	Industrial Design Overview	PF-MODENERGY-002 v2.0

> All renders are conceptual industrial-design visualisations. They communicate design and installation intent only — not manufacturing models, dimensioned drawings or certified product specifications.

Product family

One standardised Energy Module is the unit of capacity. Power Core (or Pod Power Unit) is the unit of power. The same module slides from portability to home backup to SME outdoor storage.

P1 Go

Portable backup

Aluminum unibody · fold-flat top handle · glass display · Precifarm wordmark · magnetic I/O · wireless pad · Type 2 EV trickle lead

Energy Module

2.56 kWh block

Recessed side handles · blind-mate power and comms · glass status strip · one module repeated everywhere

P2 Home

Home backup

Power Core above · four module bays · retractable handle · rear castors · essential-load sub-board ready

Precifarm Pod

SME / outdoor

Power Unit · six-module rack · service door open · sun-shield canopy · plinth and cable entries

Essential parameters

Engineering targets from PF-MODENERGY-002. Usable AC = 90 % SOC window × 94 % conversion. Not certified product specifications.

Energy Module — building block

PARAMETER	VALUE
Nominal energy	2.56 kWh
Voltage / capacity	51.2 V · 50 Ah
Chemistry	$LiFePO_4$ (LFP)
Usable AC (backup)	2.17 kWh / module
Daily cycling AC	1.68 kWh / module
Max discharge	50 A · 1.0 C
Max charge	30 A · 0.6 C
Round-trip AC→AC	90 %
Backup reserve (default)	20 % SOC

Kenya design basis

PARAMETER	VALUE
Grid	230 V · 50 Hz · Type G
Design solar (Nairobi)	4.7 PSH · 3.5 kWh/kWp/day yield
Grid cost (all-in)	KSh 20–28 / kWh
Domestic net-metering cap	10 kW
Avg. monthly outage	~8.4 h (reported)
PV performance ratio	0.75 planning · clean panels matter

Product lineup — energy vs power scale separately

PRODUCT	ENERGY	POWER	TYPICAL USE
P1 Go	~1.0 kWh	1.0 kW	Portable · lights, router, fan
P2–2.4	2.56 → 10.24 kWh	2.4 kW	Home backup · 1–4 modules
P2–4.8	5.12 → 10.24 kWh	4.8 kW	Home + pump · 2–4 modules
Pod home	5.12 → 10.24 kWh	4.5–8 kW	Installed · solar + backup
Pod SME	20.48 → 51.2 kWh	15–30 kW	Retail · cold chain

Nairobi sizing scenarios

ID	PROFILE	DAILY LOAD kWh/d	PEAK kW	MODULES	PRODUCT	SOLAR kWp
A	Essential backup	1.40	0.21	1	P1 Go / P2–2.4	0.6
B	Urban home	3.77	2.61	3	P2–4.8	1.6
C	Home + EV	11.0	6.31	8	Pod	4.1

Scenario A — essential loads

LOAD	POWER W	ENERGY Wh/d
LED lighting (8 × 9 W)	72	360
Wi-Fi router	10	240
Phone charging (2×)	20	60
Television 32"	50	200
Fan	45	180
Security / gate standby	15	360
Daily total (Scenario A)	–	**1,400**

Backup duration — usable AC energy

LOAD	1 MODULE hours	2 MODULES hours	4 MODULES hours
500 W	4.3 h	8.7 h	17.3 h
1 kW	2.2 h	4.3 h	8.7 h
2 kW	1.1 h	2.2 h	4.3 h

Key sizing rules P1 Go ≈ 77 % of one essential night · P2–2.4 (1 module) ≈ 2 nights · Scenario B needs 3 modules with reserve. Quote *expected usable* kWh, not nameplate. Full math: PF-MODENERGY-002 §16.

Designed for real sites

Installation context is part of the design brief — utility corners, SME refrigeration continuity, and solar-to-EV loops on Nairobi suburban homes.

Urban home

P2 beside consumer DB — tiled utility corner, spare module storage, living-space acceptable finish

SME retail

Wall-shaded Pod for cold storage continuity — plinth mount, filtered louvres, rooftop PV

Solar + EV

Carport charging from customer solar surplus — Phase A home energy loop, Nairobi suburban context

Design language

- Seamless brushed aluminum and matte white enclosures
- Fold-flat top handle and tilt stand (P1 Go)
- Living-space acceptable finish for urban utility rooms
- Glass UI strips — energy remaining, health, expansion
- Blind-mate module connectors — field service without rack tools
- Wall-shaded Pod placement with plinth and filtered airflow

EV charging we design & operate

From portable Spark charger and Pulse home wallbox through Depot fleet AC to Corridor DC on intercity routes — CCS2 / Type 2 for Kenya. Unified industrial design: aluminum, glass UI, Precifarm wordmark on every product.

Spark · Pulse · Pod · Boda Hub · Depot · Corridor — one visual language from boot to highway.

Spark charger
3.3 kW AC · Portable

Pulse charger
7 kW AC · Home wallbox

Corridor DC
120 kW+ · Highway CCS2

Depot station
22 kW AC · Fleet yard

Boda Hub
Swap · <5 min

Pod storage
5–10 kWh · + home charger

Precifarm EV charging hardware

PRODUCT	POWER	CONNECTOR	ROLE	TYPICAL SESSION
Spark charger	3.3 kW	Type 2	Portable top-up	~180 min
Pulse charger	7 kW	Type 2	Home overnight	~90 min
Pod + charger	7 kW + storage	Type 2	Weak-grid home	5–10 kWh
Depot station	22 kW	Type 2	Fleet / campus	~120 min
Corridor DC	120–320 kW	CCS2	Highway / hub	~60 kWh / 30 min
Boda Hub	Swap	Pack swap	E-motorcycle	<5 min
Route hub	120–320 kW	CCS2 + BESS	Reserved bus	120 kWh / 30 min

Fast DC benchmark — Tesla vs BYD vs Precifarm [BENCHMARK]

DIMENSION	TESLA SUPERCHARGER	BYD MEGAWATT FLASH	PRECIFARM CORRIDOR
Peak power (post / vehicle)	500 kW	1,000 kW	120–320 kW
Shared cabinet	1.2 MW · 8 posts	1,360 kW terminal	Modular CCS2
Previous generation	250 kW (V3)	Dual-gun GB/T boost	120 kW town class
DC voltage	180–1,000 V	1,000 V platform	200–920 V CCS2
Connector	NACS / CCS2	GB/T / MW gun	CCS2 + Type 2
Primary use	Premium car network	China flash charge	Kenya bus + intercity
In Kenya today	No network	No network	Designed · route hubs

Modular energy × EV integration (PF-MODENERGY-002 §15)

CASE	POWER / ENERGY	HARDWARE	STATUS
A · AC from solar surplus	3.7 kW	Pod / P2 + Pulse	Type 2 · near-term
P1 emergency trickle	1.0 kW max	P1 Go + lead	Not daily EV use
C · Home + EV load	6.3 kW peak	8-module Pod	2.6 kW home + 3.7 kW EV
40 km driving / day	7.2 kWh	0.18 kWh/km	~2 h @ 3.7 kW
B · EV visibility	—	Energy OS	FUTURE
C/D · V2H / V2G	Bidirectional	Pod + approved EV	FUTURE

Benchmark read (Aug 2026) Tesla V4 reaches 500 kW per post; BYD targets 1 MW at 1,000 V in China. Standard CCS2 remains ~500 A class. Precifarm targets **120–320 kW CCS2** corridor hubs with solar/BESS, reserved bus windows and M-Pesa — right-sized for Kenya, not the megawatt race. Sources: §18–§19 · Tesla Supercharger for Business · BYD Super e-Platform.

Precifarm Modular Energy + EV Charging — Design Overview
PF-MODENERGY-DESIGN-001 · v1.5 · 30 August 2026 · Engineering package: precifarm-solar-charger-stations-engineering.pdf
Corridor DC and route hubs are designed, not universally commissioned. No megawatt or V2G claims without hardware and regulatory basis.